Abstract

Due to the presence of different kinds of softwares for high level image processing, tampering of original digital images is easily done by end users. Therefore this becomes research problem as one cannot know whether present digital image is original or forged. Common type of forgery used with digital images is Region Duplication Forgery. In this dissertation we are presenting new hybrid method for Region Duplication Forgery Detection in Digital Images Based on 2D-Discrete Wavelet Transform (DWT) and Singular Value Decomposition (SVD) with goal of improving the performances in terms of hit rate, miss rate and false detection rate. Previously only 2D-DWT based method was presented, however we are further extending this method by using SVD with it so that accurate forgery area can be localized. In proposed method, each layer of RGB image is extracted and then 2D-DWT is performed on forged image. It then divides LL sub-band into overlapping blocks. Perform SVD on each block to get the dominant feature. Sort blocks based on dominant feature into bucket groups.

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Chapter 1

Introduction

Images and videos have become the main information carriers in the digital era. The expressive potential of visual media and the ease in their acquisition, distribution and storage is such that they are more and more exploited to convey information, even sensible. As a consequence, today images and videos represent a common source of evidence, both in every-day life controversies and in trials. The simplest video in TV news is commonly accepted as a certification of the truthfulness of the reported news. In a similar way, video surveillance recordings can constitute fundamental probationary material in a court of law. Together with undoubted benefits, the accessibility of digital visual media brings a major drawback. Image processing experts can easily access and modify image content, and therefore its meaning, without leaving visually detectable traces. Moreover, with the spread of low-cost, user friendly editing tools, the art of tampering and counterfeiting visual content is no more restricted to experts. As a consequence, the modification of images for malicious purposes is now more common than ever. Digital Image Forensics is that branch of multimedia security that, together with Digital Watermarking, aims at contrasting and exposing malicious image manipulation.

1.1 Digital Image Forensics

The three major determinations of Digital Image Forensics are; Detection, source identification of digital Image, and Image Forgery Detection. The Detection of Computer Generated Image

is very hard and difficult because of its high quality and high realism realistic make confusion for viewers to judge reality. In image source, identification the digital device determines like cameras, and scanners. Every device have distinctive footprint of image. In this type of forensics two approaches use in first method the image source identification attempt to resolve the source image class properties, and second method identify the image potential source device through novel hardware characteristics. In last, the Forgery Detection is an intrusting method for digital image forgery. The forgery detection has algorithms and techniques for detection and traces image tampering. The traces are used for proof of tampering [3].

1.2 History

The driving force behind the digital image forensics is image forgery. Image forgery can be traced back to as early as 1840s when Hippolyte Bayrad created the very first fake image 1.1, in which he was shown committing a suicide [1].
In 1860s another fake image appeared in which the head of Abraham Lincoln (the then US President) was fixed over the body of an adversary politician, John Calhoun [1].
Many other instances could be found in the history of the period spanning over the late 19th century and the early 20th century. With the advent of the Hollywood factor, motion films with synthesised scenes have become a norm. Various tampered war photographs then appeared and the use of such photographs was considered part and parcel of a successful war propaganda. Computers revolutionised the art with early super examples of the work as Terminator, Jurassic Park and Forest Gump, in the early 1990s. The 21st century dawned with the tragedy of 9/11, when many innocent civilians lost their lives. Different videos of Osama bin Laden emerged, of which many were, later on, determined to be fake through forensics analyses. With the passage of time, many advanced software editing tools have been made available to the end users who can now easily alter an image with little or no effort. With such a long history of forgeries, photography, especially digital photography has lost its innocence. Over the past few years, many techniques have been introduced to tamper images or videos [1].



Figure 1.1: First Fake Image

1.3 Areas of application

- Authentication of images captured from CCD (charge coupled device) cameras
- Authentication of information available in an image
- Authenticity of evidences
- Fingerprint recognition
- Document authentication

Chapter 2

Literature Review

2.1 Classification of Image Authentication Techniques

Forgery detection intends to verify the authenticity of images. For authentication of images several methods have been developed. In this paper we broadly classify these methods into two classes: Active authentication and Passive authentication [5]. The classification is based on the fact whether the original image is available or not. Under each class the methods are further sub divided. The hierarchy is shown in figure 2.1.

2.1.1 Active Authentication

In active authentication techniques prior information about the image is indispensable to the process of authentication. It is concerned with data hiding where some code is embedded into the image at the time of generation. Verifying this code authenticates the originality of image [5]. Active authentication methods are further classified into two types:

1. Digital watermarking
   Digital water marks are embedded into the images at the time of image acquisition or in processing stage.

2. Digital signatures
   Digital signatures embed some secondary information, usually extracted from image, at

the acquisition end into the image .A lot of work has been carried in both digital signatures and digital watermarking. The main drawback of these approaches remains that they are to be inserted into the images at the time of recording using special equipments thus prior information about image becomes indispensable.

2.1.2 Passive Authentication

Passive authentication also called image forensics is the process of authenticating images with no requirement of prior information just the image itself. Passive techniques are based on the assumption that even though tampering may not leave any visual trace but they are likely to alter the underlying statistics. It is these inconsistencies that are used to detect the tampering. Exhaustive research survey has been carried out in this field of passive image forensics. Passive techniques are further classified as forgery dependent methods and forgery independent methods [5].
Forgery dependent detection methods are designed to detect only certain type of forgeries such as region duplication or copy-move and splicing which are dependent on the type of forgery carried out on the image while as forgery independent methods detect forgeries independent of forgery type but based on artifact traces left during process of re-sampling and due to lighting inconsistencies.

1. Region Duplication or Copy-Move Image Forgery
   In Region duplication forgery (or copy move or cloning), part of the image of any size and shape is copied and pasted to another location in the same image to hide some important information or to duplicate portions of the image as shown in figure 2.2. As the copied part came from the same image, its important properties such as noise, color and texture do not change and make the detection process difficult [2].

2. Image Forgery using Splicing
   Image splicing uses cut-and-paste techniques from one or more images to produce a new fake image. When splicing is performed carefully, the borders between the spliced regions can visually be imperceptible. Splicing, however, disturbs the high order Fourier statistics such as the bispectrum; these statistics can subsequently be used in detecting

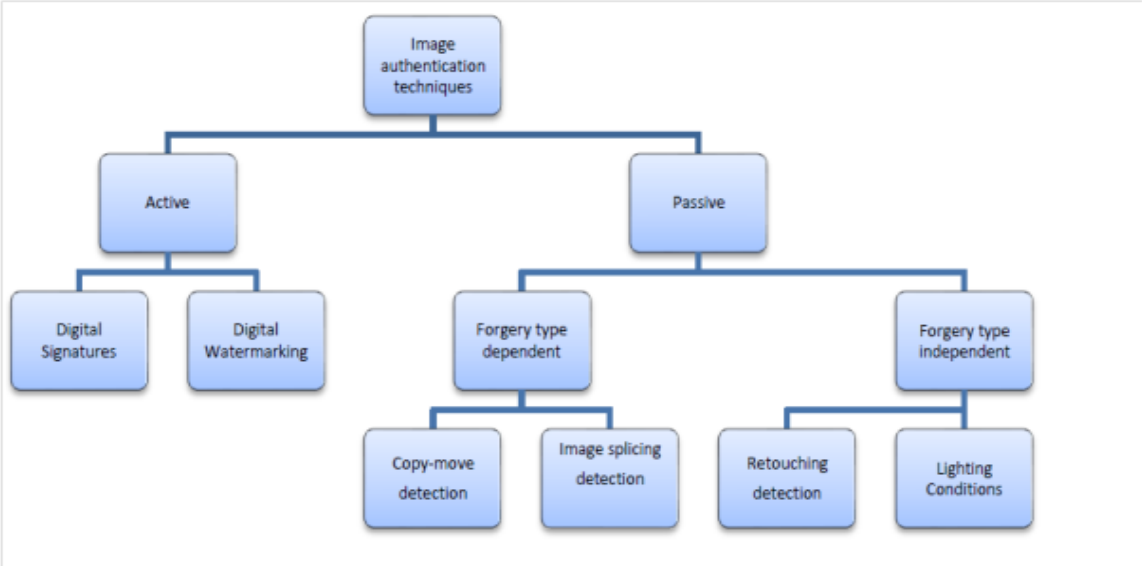


Figure 2.1: Classification of Image Authentication Techniques



Figure 2.2: An example of region duplication copy-move forgery; (left) original image (right) forged image

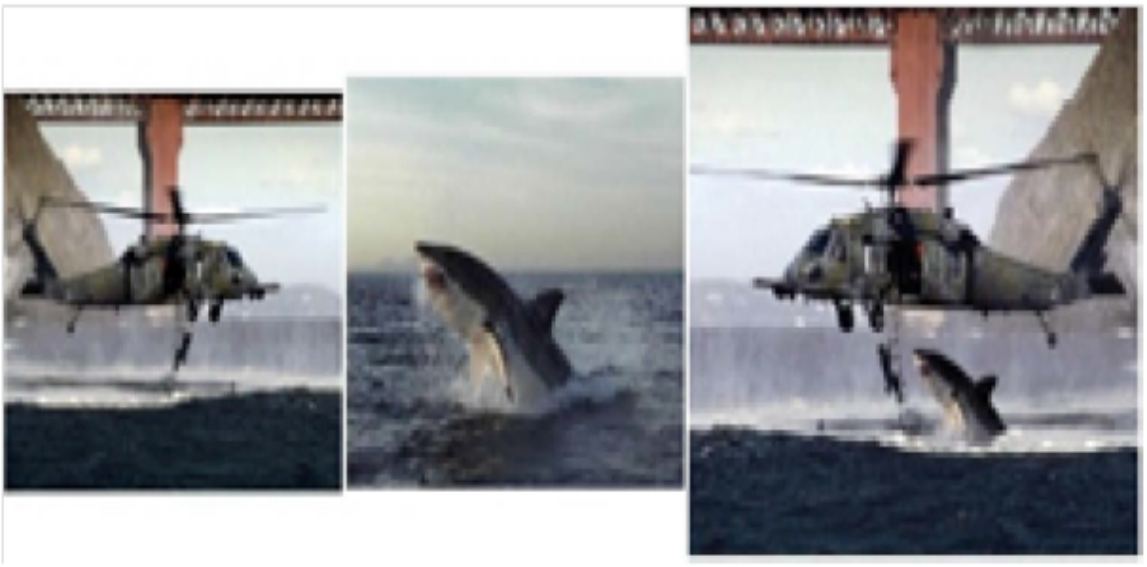


Figure 2.3: Example of image splicing



Figure 2.4: Example of Image Resampling or Retouching

forgery. Figure 2.3, shows a nice example of image splicing in which the images of the shark and the helicopter are merged into one image [2].

3. Image Resampling or Retouching
   To create a high quality forged image, some selected regions have to undergo geometric transformations like rotation, scaling, stretching, skewing, flipping etc. The interpolation step plays a central role in the resampling process and introduces non-negligible statistical changes. Resampling introduces specific periodic correlations into the image. These correlations can be used to detect forgery caused by resampling. In figure 2.4, the image on the left is the original scene while the one on the right is the forged image obtained by rotation and scaling it [2].

2.2 Region Duplication Forgery

Region duplication forgery is one of the most popular forms of tampering in which some region is copied from a particular location in an image and thereafter pasted at one or more locations within the same image or a different image of preferably the same scene.
Region duplication detection or copy move detection methods are either keypoint-based methods or block-based methods. In both cases, preprocessing of the images is possible. For instance, most methods operate on grayscale images, and as such require that the color channels be first merged. The common processing pipeline for detection of region duplication or copy move forgery detection [4] is shown in figure 2.5.

1. **Feature Extraction**
   For feature extraction, block-based methods subdivide the image in rectangular regions. For every such region, a feature vector is computed. Similar feature vectors are subsequently matched. By contrast, keypoint-based methods compute their features only on image regions with high entropy, without any image subdivision. Similar features within an image are afterwards matched. A forgery shall be reported if regions of such matches cluster into larger areas. Both, keypoint and block-based methods include further filtering for removing spurious matches. An optional postprocessing step of the detected

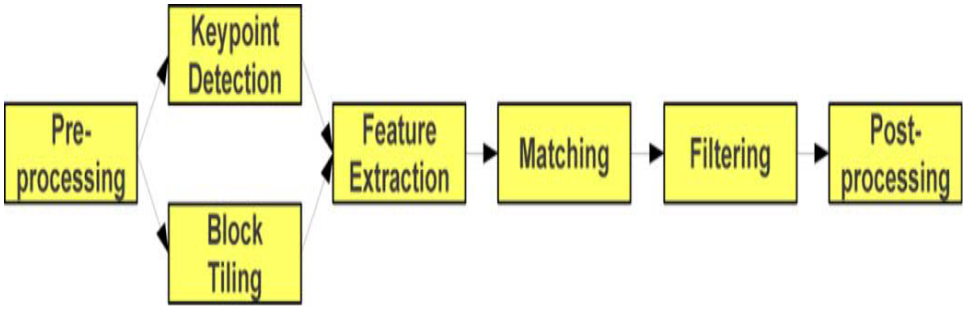


Figure 2.5: Common processing pipeline for detection of region duplication or copy move forgeries

regions may also be performed, in order to group matches that jointly follow a transformation pattern [4].

2. **Matching**
   High similarity between two feature descriptors is interpreted as a cue for a duplicated region. For block based methods, lexicographic sorting is used in identifying similar feature vectors. In lexicographic sorting a matrix of feature vectors is built so that every feature vector becomes a row in the matrix. This matrix is then row-wise sorted. Thus, the most similar features appear in consecutive rows. Matching with a kd-tree yields a relatively efficient nearest neighbor search. Typically, the Euclidean distance is used as a similarity measure. The use of kd-tree matching leads, in general, to better results than lexicographic sorting, but the memory requirements are significantly higher. However, that the dimensions of some feature sets are ordered by importance. For these features, the performance gain over lexicographic sorting is minimal [4].

3. **Filtering**
   Filtering schemes have been proposed in order to reduce the probability of false matches. For instance, a common noise suppression measure involves the removal of matches between spatially close regions. Neighboring pixels often have similar intensities, which

can lead to false forgery detection. Different distance criteria were also proposed in order to filter out weak matches. For example, several authors proposed the Euclidean distance between matched feature vectors. In contrast, some authors proposed the correlation coefficient between two feature vectors as a similarity criterion [4].

4. **Postprocessing**
   The goal of this last step is to only preserve matches that exhibit a common behavior. Consider a set of matches that belongs to a copied region. These matches are expected to be spatially close to each other in both the source and the target blocks (or keypoints). Furthermore, matches that originate from the same copy-move action should exhibit similar amounts of translation, scaling and rotation. The most widely used postprocessing variant handles outliers by imposing a minimum number of similar shift vectors between matches. A shift vector contains the translation (in image coordinates) between two matched feature vectors.

2.2.1 Region Duplication Forgery Detection Techniques

1. **Block-based methods**
   Many techniques are based on block-based scheme where image separated into small overlapping or non- overlapping blocks. Then the separated blocks compare with each other for analyzing the matching areas and these areas cover through matching blocks [3]. There are types of block based methods such as Moment-based Techniques which includes Blur Movement, Hu movement, SIFT movement; Intensity based methods; and Dimensionality reduction-based Methods which includes

   - PCA(Principal Component Analysis)
     Principal Component Analysis (PCA) is known as the best data representation in the least-square sense for classical recognition. It is commonly used to reduce the dimensionality of images and retain most information.
   - DCT(Discrete Cosine Transform)
     Many video and image compression algorithms apply the discrete cosine transform (DCT) to transform an image to the frequency domain and perform quantization

for data compression]. One of its advantages is the energy compaction property; that is, the signal energy is concentrated on a few components while most other components are zero or negligibly small. This helps separate an image into parts (or spectral sub-bands) of hierarchical importance (with respect to the images visual quality). The popular JPEG compression technology uses the DCT to compress an image.

- Singular Value Decomposition (SVD)
  It is robust to post-processing techniques and computationally is less complex.
- DWT (Discrete Wavelet Transform) The DWT provides a time-frequency representation of the signal. It uses multi-resolution technique by which different frequencies are analyzed with different resolutions. DWT is a linear transform that operates on vector whose length is an integer multiple of 2 while separating the data vector into different frequency components.
- KPCA (Kernel Principal Component Analysis) KPCA is a nonlinear PCA developed by using the kernel method. KPCA maps the original inputs into a high dimensional feature space using a kernel method. KPCA is computationally intensive and takes a lot more time compared to PCA. The reason being that the number of training data points in KPCA is much higher then PCA. So number of principle components that need to be estimated is also much larger.

2. **Key point-based Methods**
   Unlike block-based algorithms, keypoint-based methods rely on the identification and selection of high-entropy image regions (i.e., the keypoints). A feature vector is then extracted per keypoint. Consequently, fewer feature vectors are estimated, resulting in reduced computational complexity of feature matching and postprocessing. The lower number of feature vectors dictates that postprocessing thresholds are also to be lower than that of block-based methods. A drawback of keypoint methods is that copied regions are often only sparsely covered by matched keypoints. If the copied regions exhibit little structure, it may happen that the region is completely missed [4]. There are two types of methods such as SIFT (Scale Invariant Feature Transform) algorithm, Speeded up Robust Features (SURF).

2.3 Related Work

Fattah, S.A.; Ullah, M.M.I.; Ahmed, M.; Ahmmed, I.; Shahnaz, C., [6] presents a scheme for copy-move forgery detection in digital images based on 2D-DWT with the help of block matching procedure. The proposed candidate block selection algorithm helps in avoiding the huge computational burden involved in block matching operation among all blocks. Moreover, in DWT coefficient matching, instead of using both approximate and detail coefficients, only the former type is employed. In the second stage, as the candidate blocks are matched with all overlapping blocks, high level of detection accuracy is achieved both in case of single and multiple copy move forgeries.

Christlein, V.; Riess, C.; Jordan, J.; Riess, C.; Angelopoulou, E., [4] aim to answer which copy-move forgery detection algorithms and processing steps (e.g.,matching, filtering, outlier detection, affine transformation estimation) perform best in various postprocessing scenarios.Their Experiments show, that the keypoint-based features Sift and Surf, as well as the block-based DCT, DWT, KPCA, PCA, and Zernike features perform very well. These feature sets exhibit the best robustness against various noise sources and downsampling, while reliably identifying the copied regions.

Salam a.thajeel, Ghazali sulong, [3]review about digital image forensics and its types. Then discuss about current research with an analysis of the existing approaches for detect image tampering, which are classified into block-based method and key point-based method. Then they demonstrate the popular techniques of two methods and analysis of several techniques. Finally, the all techniques advantages and disadvantages are summarized and some future research directions were pointed out.

Saba Mushtaq, Ajaz Hussain Mir, [5]attempts to provide an overview of various digital image forgeries and the state of art passive methods to authenticate digital images.

Er. Saiqa Khan,Er. Arun Kulkarni, [13] describes blind forensics approach for detecting Copy-Move forgery. This technique works by first applying DWT (Discrete Wavelet Trans-

form) to the input image to yield a reduced dimension representation. Then the compressed image is divided into overlapping blocks. These blocks are then sorted and duplicated blocks are identified using Phase Correlation as similarity criterion. Due to DWT usage, detection is first carried out on lowest level image representation. This approach drastically reduces the time needed for the detection process.

G. Li, Q. Wu, D. Tu and S.J. Sun, [7] proposed a blind forensics approach based on DWT (Discrete Wavelet Transform) and SVD (Singular Value Decomposition) to detect the specific artifact. Firstly, DWT is applied to the image, and SVD is used on fixed-size blocks of low-frequency component in wavelet sub-band to yield a reduced dimension representation. Then the SV vectors are then lexicographically sorted and duplicated image blocks will be close in the sorted list, and therefore will be compared during the detection steps. The experimental results demonstrate that the proposed approach can not only decrease computational complexity, but also localize the duplicated regions accurately even when the image was highly compressed or edge processed.

Feng Liu1, and Hao Feng, [11] presents an efficient algorithm for image copy-move forgery detection and localization based on DWT and SVD. Experiment results demonstrate that this proposed algorithm can effectively detect multiple copy-move forgery and precisely locate the duplicated regions, even when an image was distorted by Gaussian blurring, JPEG compression and their mixed operations.

2.4 Motivation

With the ease of image tampering and counterfeiting visual content, which is known as image forgery, by using the photo editing software, authenticity of digital images becomes a crucial issue in several applications, such as news, legal documents, and scientific innovations. It becomes a severe threat to security that anyone may access and modify the content of an image without leaving visually detectable traces. Among different types of image forgery, image cloning also known as copy-move image forgery is considered as very harmful way of tam-

pering images. One can change the whole meaning of an image as in this type of forgery a portion of a certain image is first copied and then pasted on that image or another one to cover unwanted portion or to present a false situation. In order to detect copy-move forgery, generally block by block similarities are searched in terms of internal noise pattern and/or spatial variation. Different capturing devices introduce different background noises in the image. In noise based methods, a region in the given image that lacks the noise pattern is considered as forged region. However, a common problem of these approaches is the requirement of a number of pictures taken by the same camera, which may not be always available. Key point based algorithms rely on identification and selection of high entropy image regions but may cause problem in detecting forged region exhibiting little structure. Block based methods reduce the computational complexity. Also selecting candidate blocks for matching procedure reduce the computational burden.

Chapter 3

Existing Region Duplication Detection Technique

3.1 Existing System [6]

In literature, a copy-move image forgery detection scheme is developed based on a block matching algorithm. In that, 2D-DWT is performed on the forged image and then DWT domain blocks are considered, where only approximate DWT coefficients are utilized. In order to reduce the computational burden, instead of performing block matching operation among all blocks, some candidate blocks are first selected from the non-overlapping blocks based on a similarity measure. In the next stage, all overlapping blocks are compared with the candidate blocks. A similarity criterion is introduced to finally detect the forged blocks. The existing system architecture is shown in figure 3.1.

3.1.1 DWT

Given suspicious or forged color image, first, three layers of data (R, G, and B) are collected and then the 2D-DWT is performed on each layer. The DWT provides a time-frequency representation of the signal. It uses multi-resolution technique

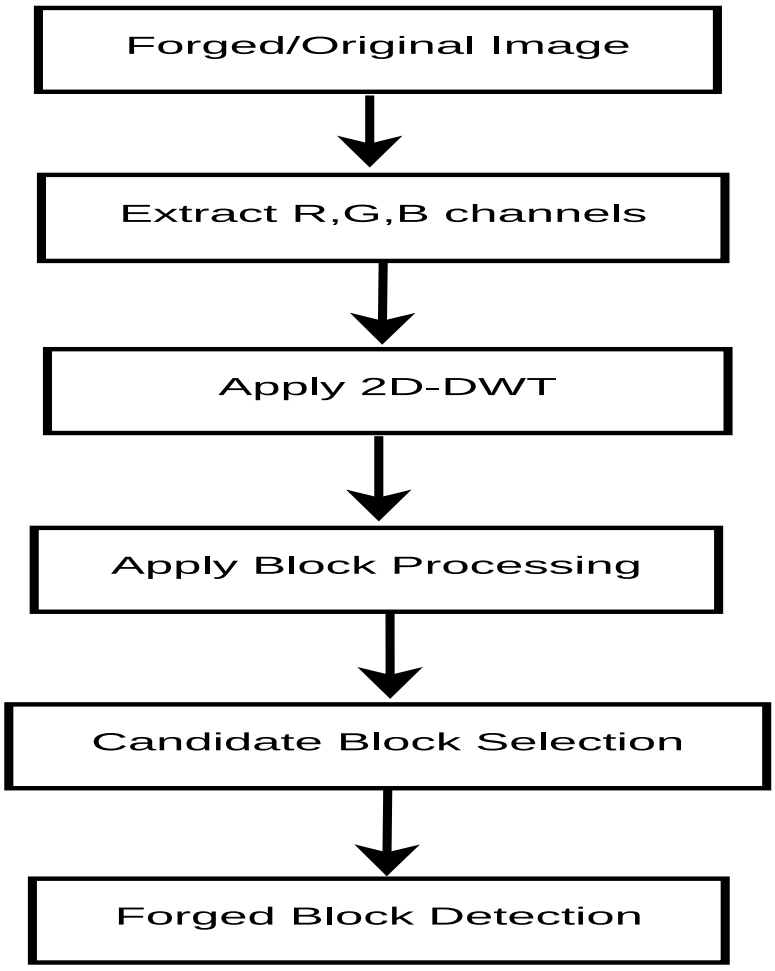


Figure 3.1: Existing System

by which different frequencies are analyzed with different resolutions. Because of its innate multi-resolution uniqueness, the DWT is employed in the proposed method. The continuous wavelet transform of a signal s(t) using a mother wavelet (t) is mathematically defined as (refer equation 3.1).

$$C(a,b) = \frac{1}{\sqrt{a}} \int s(t)\psi(\frac{t-b}{a})dt \tag{3.1}$$

where a and b represent scale and shift parameters, respectively. The DWT coefficients are obtained by restricting the scale a to powers of 2 and the position b to integer multiples of the scales, and are given by

$$c_{j,k} = 2^{\frac{j}{2}} \int s(t)\psi(2^j t - k)dt \tag{3.2}$$

where j and k are integers and j,k are orthogonal baby wavelets defined as

$$\psi_{j,k} = 2^{\frac{j}{2}}\psi(2^j t - k) \tag{3.3}$$

One advantage of using the DWT with decimation is that the size of the transformed data will be reduced to one half at each level. However, in this paper, only level one decomposition is used, which results in four decomposed sub images: LL, LH, HL, and HH. The HL band indicates the variation along X axis while the LH band indicates the Y-axis variation. The power is more compact in the LL band that corresponds to the approximation image and this LL band image is used in the proposed scheme.

3.1.2 Candidate Block Selection

The proposed method is based on block by block matching principle. The LL band DWT transformed image is divided into overlapping blocks. In the proposed scheme, square blocks are used for convenience and one pixel shifting is employed to obtain overlapping blocks. However, effect of changing block size and area of overlaps is investigated in detail. It is observed that too large block size, although provide a large number of pixels which is beneficial in terms of statistical similarity

measure, it may result in misleading information in case of small sized forgeries. On the other hand, too small block size will unnecessarily increase the computational burden. Hence, a moderate block size is preferred. On the other hand, overlapping only by one pixel is considered in order to avoid chances of missing similar regions. The idea of block-matching procedure is to extract features from each block and then to find the duplicated blocks based on the similarity in their features. In this regard, like conventional methods, if all blocks are taken into consideration in the matching procedure, it will cause huge computational burden. At the first stage, we propose to select a few candidate blocks representing a high level of similarity, which will next be used to compare with all overlapping blocks. In this way, comparison among all overlapping blocks will not be necessary. It is to be mentioned that among all overlapping blocks, there are a small number of nonoverlapping blocks which will be sufficient to consider for determining possible candidates. Hence, at the first stage, similarity measure is carried out among only the nonoverlapping blocks. Before measuring similarity all nonoverlapping blocks are lexicographically sorted. We propose to use correlation coefficient as a similarity metric. In the calculation of correlation coefficient, 2D representation of wavelet coefficients of a block is transformed into a vector. A pair of nonoverlapping blocks showing high value of correlation coefficient is considered as probable candidate.

Similarity search is carried out for every possible pair of nonoverlapping blocks for each layer. Block-pairs possessing value of correlation coefficient above a threshold value (CT) (for all three layers) are taken as candidate blocks (without any repetition). It is obvious that selecting a higher threshold will provide a small number of candidate blocks, which will increase the risk of missing the desired forged blocks. On the other hand, lower threshold value will unnecessarily provide a large number of candidate blocks, which will increase the risk of inclusion of blocks with natural similarity (not the similarity due to forgery) and eventually it will increase the computational burden. Hence, a reasonably higher threshold value is chosen.

3.1.3 Forged Block Detection

Once the candidate blocks are obtained, similarity between a candidate block and all overlapping blocks need to be measured. It is to be noted that in case of non-overlapping blocks, the intra-block distance is higher in comparison to that of the overlapping blocks. Hence, correlation coefficient serves as a potential similarity measure. On the other hand, intra block distance is very small for overlapping blocks and it is likely to obtain quite similar values of correlation coefficient in many cases. As an alternate, Euclidean distance based similarity measure is employed, this offers more consistent results. In the distance calculation, 2D representation of wavelet coefficients of a block is transformed into a vector. Here also a threshold value (CD) of distance is chosen and average of three (for three layers) block-distances below CD are selected as the forged blocks.

3.2 Limitations of Existing System

From the literature,region duplication forgery detection techniques we came to know that traditional block based approach has some problems described below:

- The False Detection Ratio(FDR) is very high for block size of 5*5.
- In region duplication forgery, if tampered image contains multiple copied regions then the false detection ratio is more.
- This method fails to detect flat region copy move image forgery.

Chapter 4

Problem Definition

We have seen the limitation of existing system in section 3.2. The proposed system should be design in such manner that, it will elevate these limitations. So, the problem definition of our proposed system should be as follows:

- To develop a system which handles Detection of multiple copy move forgery.
- To develop an algorithm which will improve FDR ratio for block size of 5*5. That is detection of smallest copied region.
- To develop an algorithm which detects flat region copy move forgery.
- To develop an algorithm which detects copy move forgery when only tampered image available with us.

Chapter 5

Proposed Work

5.1 Proposed System

Detection of the copy-move forgery done on the same image is very challenging due to the fact that it is visually difficult to trace out the copied region if it is done carefully. In this proposed system, a combined approach of 2D-Discrete Wavelet Transform(2D-DWT) and Singular Value Decomposition(SVD) for copy-move forgery detection is presented. In the proposed method, all three layers of color image are utilized. The figure 5.1 illustrates the proposed system architecture.

5.1.1 Database

For forgery detection performance of the proposed algorithm, we are using several images taken from publicly available databases such as MICC F220 and MICC 2000.For the purpose of testing, forged images are prepared by copying certain portion of the image and pasting it to another portion of that image and then saving the forged image in a standard format (JPEG). At the beginning of the testing phase, it is assumed that only the forged image in a standard format (JPEG) is available.

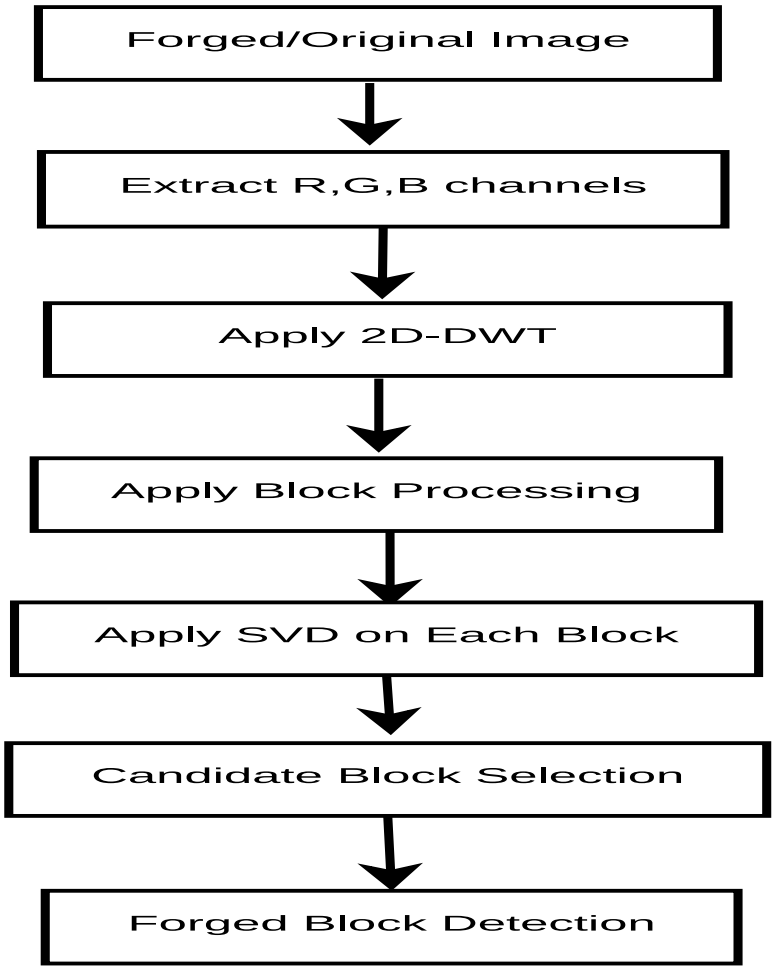


Figure 5.1: Proposed System

5.1.2 2D-Discrete Wavelet Transform(2D-DWT)

An image is represented as a two-dimentional array of coefficients, each coefficient representing the brightness level in that point. When looking from a higher perspective, we can't differentiate between coefficients as more important ones, and lesser important ones. But thinking more intuitively, we can. Most natural images have smooth colour variations, with the fine details being represented as sharp edges in between the smooth variations. Technically, the smooth variations in colour can be termed as low frequency variations and the sharp variations as high frequency variations.

The low frequency components (smooth variations) constitute the base of an image, and the high frequency components (the edges which give the detail) add upon them to refine the image, thereby giving a detailed image. Hence, the smooth variations are demanding more importance than the details.

Separating the smooth variations and details of the image can be done in many ways. One such way is the decomposition of the image using a Discrete Wavelet Transform (DWT).

The procedure goes like this. A low pass filter and a high pass filter are chosen, such that they exactly halve the frequency range between themselves. This filter pair is called the Analysis Filter pair. First, the low pass filter is applied for each row of data, thereby getting the low frequency components of the row. But since the lpf is a half band filter, the output data contains frequencies only in the first half of the original frequency range. So, by Shannon's Sampling Theorem, they can be subsampled by two, so that the output data now contains only half the original number of samples. Now, the high pass filter is applied for the same row of data, and similarly the high pass components are separated, and placed by the side of the low pass components. This procedure is done for all rows.

Next, the filtering is done for each column of the intermediate data. The resulting two-dimensional array of coefficients contains four bands of data, each labelled as LL (low-low), HL (high-low), LH (low-high) and HH (high-high). The LL band can be decomposed once again in the same manner, thereby producing even more subbands. This can be done upto any level, thereby resulting in a pyramidal decomposition as shown figure 5.2. The LL band at the highest level can be classified as most important, and the other 'detail' bands can be classified as of lesser importance, with the degree of importance decreasing from the top of the pyramid

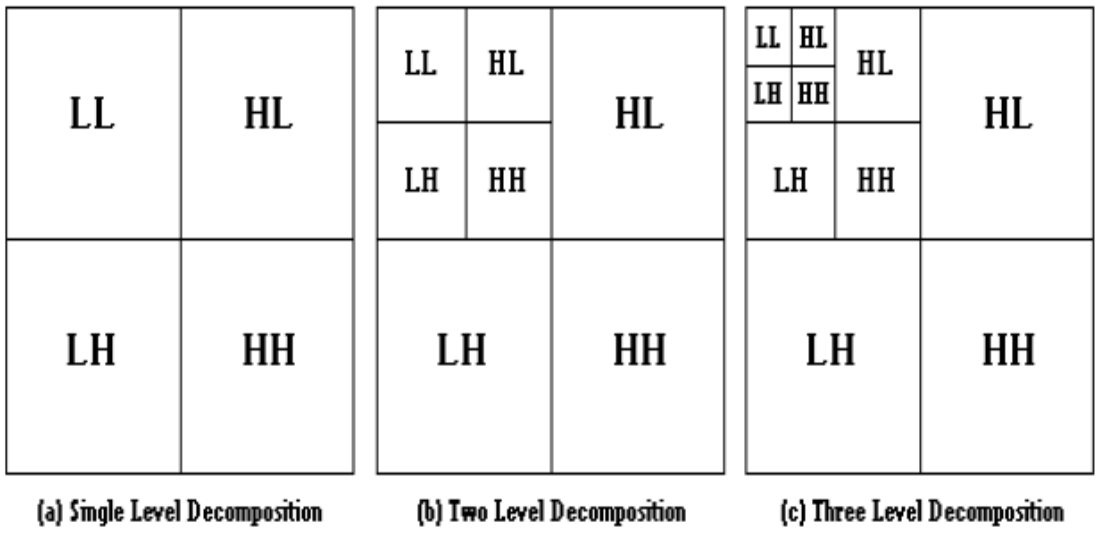


Figure 5.2: Pyramidal Decomposition of an Image

to the bands at the bottom.

5.1.3 Singular Value Decomposition (SVD)

One of the basic and most important tools of modem numerical analysis, particularly numerical linear algebra, is the singular value decomposition. The SVD was established for real square matrices in the 1870's by Beltrami and Jordan. Singular value decomposition (SVD) is a matrix factorization and has three properties, namely, stability, scaling property and rotation invariance, which represents algebraic and geometric invariant properties of an image. SVD has been used in a large amount of fields such as signal processing, data compression and pattern analysis [11]. SVD is not only an important tool for exploratory data analysis, dimensionality reduction and data compression, but also a method for noise reduction. The singular values are unique for a matrix, which form a steady representation of image blocks. SVD Definition: any real m x n (In general, let $m \geq n$) matrix A can be decomposed uniquely as

$$A = U \wedge V^T \tag{5.1}$$

Where U is an m X m orthogonal matrix, V is an n x n orthogonal matrix, and A is an m x n matrix whose off-diagonal entries are all zeros and whose diagonal elements satisfy

$$\sigma_1 \geq \sigma_2 \geq ... \geq \sigma_n \geq 0 \tag{5.2}$$

It can be shown that r = rank(A) equals the number of nonzero singular values. σ_i, i =l,...r represent the SVs in descending order [7]. There are two purposes for using SVD:

1. SV vector is the unique, steady representation of a block. It is optimal for given image in the sense that the energy packed in a given number of transformation coefficients is maximized.

2. It further reduces feature dimension from m x n to r.

5.2 Proposed Algorithm

Step 1: Read colour forged image
Step 2: Extract R, G, and B channels
Step 3: Apply pre-processing on each channel to reduce noise
Step 4: Apply 2D-DWT on each channel and extract its approximation coefficient for further processing
Step 5: On each approximation coefficient of every channel, overlapping blocks of size b*b are created and then apply SVD to get new feature arrays of overlapping blocks
Step 6: On each layer, every overlapping blocks are transformed into row matrix
Step 7: Extract rows those are representing non-overlapping blocks from row matrix
Step 8: Correlation coefficient using Euclidian distance between pair of non-overlapping rows
Step 9: Select threshold Ct of correlation coefficient in order to find blocks whose value of correlation coefficient is more than set threshold value Ct
Step 10: The extract blocks using Ct are treated as candidate blocks
Step 11: Further distance is computed between every candidate rows against all other overlapping blocks
Step 12: three different distances obtained on three channels; hence threshold value Cd is selected for extracting forged blocks. The blocks whose value is below Cd are considered as

forged block in original image

5.3 Evaluation Parameters

At the beginning of the testing phase, it is assumed that only the forged image in a standard format (JPEG) is available. For performance evaluation, we are considering most widely used parameters, hit rate, miss rate, and false detection rate (FDR), which can be defined as

$$HitRate = \frac{ImageDetectedasForgedbeingForged}{Forgedimage} * 100 \tag{5.3}$$

$$MissRate = \frac{ImageDetectedasNotForgedbeingForged}{Forgedimage} * 100 \tag{5.4}$$

$$FDR = \frac{ImageDetectedasForgedbeingOriginal}{Forgedimage} * 100 \tag{5.5}$$

Chapter 6

Conclusion

The region duplication image forgery detection plays an important role in digital image forensics. The proposed algorithm presents a combined method of 2D-DWT and SVD for region duplication image forgery detection. The proposed candidate block selection algorithm will help in avoiding the huge computational burden involved in block matching operation among all blocks. In this proposed method we will apply SVD for feature extraction, which will help in candidate block selection. SVD will improve the accuracy of region duplication image forgery detection in both case of single and multiple region duplication forgeries. The proposed algorithm will improve performance in terms of hit rate, miss rate, and false detection rate.

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